Exercises week 6 – Answers

8CM10 - Introduction to Quantum Physics and Structure of Matter

Electron Configurations & Quantum Numbers

1.

a) List the six possible sets of quantum numbers (n, l, m_l, m_s) of a $2p$ electron.
b) Give the electron configuration of ${}^{12}_{6}C$ (carbon).
c) Suppose we have an atom such as carbon, which has two $2p$ electrons. Ignoring the Pauli principle, how many different possible combinations of quantum numbers of the two $2p$ electrons do exist?
d) How many of the possible combinations of part b) are eliminated by applying the Pauli principle?
e) Suppose carbon is in an excited state with configuration $2p^1 3p^1$. Does the Pauli principle restrict the choice of quantum numbers for the electrons? How many different sets of quantum numbers are possible for the two electrons?

a) All 6 combinations with $n = 2, l = 1, m_l = -1, 0, 1$ and $m_s = -\frac{1}{2}, +\frac{1}{2}$:

$$\begin{array}{llll} n=2, & l=1, & m_l=-1, & m_s=-\frac{1}{2} \\ n=2, & l=1, & m_l=-1, & m_s=\frac{1}{2} \\ n=2, & l=1, & m_l=0, & m_s=-\frac{1}{2} \\ n=2, & l=1, & m_l=0, & m_s=\frac{1}{2} \\ n=2, & l=1, & m_l=1, & m_s=-\frac{1}{2} \\ n=2, & l=1, & m_l=1, & m_s=\frac{1}{2} \end{array}$$

b) Carbon (C) has (in the ground state) electron configuration: $1s^2\ 2s^2\ 2p^2$.

c) In the p shell ($l = 1$) are 6 positions, so (without Pauli principle) we have $6 \times 6 = 36$ combinations.

d) The 6 combinations with the same quantum numbers are forbidden by the Pauli principle. Hence 36-6=30 combinations are possible.

e) Now there are no restrictions since both electrons have already a different value for n. Hence there are 36 sets of quantum numbers possible.

2. Nitrogen ($Z = 7$) has three electrons in the $2p$ level (in addition to two electrons each in the $1s$ and $2s$ levels).
   a) Consistent with the Pauli principle, what is the maximum possible value of the total m_s of all seven electrons?
   b) List the quantum numbers of the three $2p$ electrons that result in the largest total m_s.
   c) If the electrons in the $2p$ level occupy states that maximize m_s, what would be the maximum possible value for the total m_l?
   d) What would be the maximum possible total ml if the three $2p$ electrons were in states that did not maximize m_s?

(a) The two electrons in the $1s$ level have m_s of +1/2 and −1/2, so they do not contribute to the total m_s, and the same is true for the two electrons in the $2s$ level. In the $2p$ level, there are three different possible values of m_l, and for each of those values we can assign a set of quantum numbers with $m_s = +1/2$, so the maximum possible value of the total m_s is +3/2.
(b) $(n, l, m_l, m_s) = (2, 1, +1, +1/2), (2, 1, 0, +1/2), (2, 1, -1, +1/2)$
(c) There is only one possible value of the total m_l in the states that maximize m_s, and from the states listed in (b) that value is $+1 + 0 + (-1) = 0$.
(d) We could maximize the total m_l by giving the first $2p$ electron $m_l = +1$, and the second electron can also have $m_l = +1$ if we give these two electrons opposite values of m_s. The third electron cannot have $m_l = +1$, so we must assign it $m_l = 0$ and the maximum total m_l is +2.

3.

a) List all elements (with $Z < 100$) having a p^3 configuration.
b) List all elements (with $Z < 100$) having a d^7 configuration.

a) Elements with p^3 configuration:

N (Z=7) : $1s^2\ 2s^2\ 2p^3$
P (Z=15) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
As (Z=33) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^3$
Sb (Z=51) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^3$
Bi (Z=83) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^6\ 6s^2\ 4f^{14}\ 5d^{10}\ 6p^3$

b) Elements with d^7 configuration:

Co (Z=27) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^7$
Rh (Z=45) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^7$
Ir (Z=77) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^6\ 6s^2\ 4f^{14}\ 5d^7$

4. Using the wavelengths given in Figure 8.4, compute the energy difference between the $3d$ and $4d$ states in lithium; do the same for sodium. Compare those values with the corresponding $n = 4$ to $n = 3$ energy difference in hydrogen. Why is the agreement so good, considering the different values of Z?

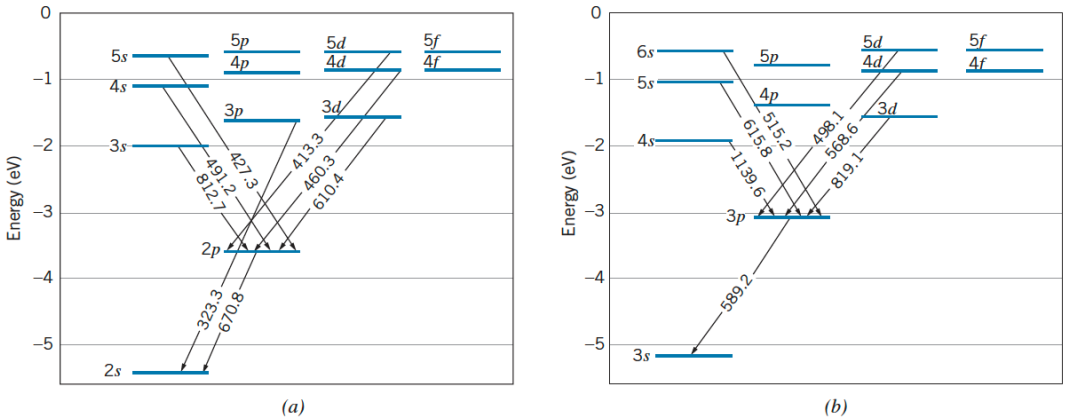


FIGURE 8.4 (*a*) Energy-level diagram of lithium, showing some of the transitions (labeled with wavelength in nm) in the optical region. (*b*) Energy-level diagram for sodium. Because of the fine-structure splitting, the 3*p* level in sodium is actually a very closely spaced pair of levels, so all transitions involving that level show two closely spaced wavelengths. The wavelength shown here is the average of the two. The fine-structure splitting is negligible for the other levels in sodium and for all of the levels in lithium.

In lithium,

$$3d\rightarrow 3p \qquad \lambda = 610.4\text{ nm} \qquad \Delta E = \frac{hc}{\lambda} = \frac{1240\text{ eV}\cdot\text{nm}}{610.4\text{ nm}} = 2.03\text{ eV}$$

$$4d\rightarrow 3p \qquad \lambda = 460.3\text{ nm} \qquad \Delta E = \frac{hc}{\lambda} = \frac{1240\text{ eV}\cdot\text{nm}}{460.3\text{ nm}} = 2.69\text{ eV}$$

$$\Delta E(4d \rightarrow 3d) = \Delta E(4d \rightarrow 3p) - \Delta E(3d \rightarrow 3p) = 2.69\text{ eV} - 2.03\text{ eV} = 0.66\text{ eV}$$

In sodium,

$$3d\rightarrow 3p \qquad \lambda = 819.1\text{ nm} \qquad \Delta E = \frac{hc}{\lambda} = \frac{1240\text{ eV}\cdot\text{nm}}{819.1\text{ nm}} = 1.51\text{ eV}$$

$$4d\rightarrow 3p \qquad \lambda = 568.6\text{ nm} \qquad \Delta E = \frac{hc}{\lambda} = \frac{1240\text{ eV}\cdot\text{nm}}{568.6\text{ nm}} = 2.18\text{ eV}$$

$$\Delta E(4d \rightarrow 3d) = \Delta E(4d \rightarrow 3p) - \Delta E(3d \rightarrow 3p) = 2.18\text{ eV} - 1.51\text{ eV} = 0.67\text{ eV}$$

In hydrogen, $\Delta E(4 \rightarrow 3) = (-13.6\text{ eV})\left(\frac{1}{4^2} - \frac{1}{3^2}\right) = 0.66\text{ eV}$

The agreement is very good, because in lithium and sodium the outer electron is screened by the $Z-1$ inner electrons, and so the outer energy levels are well approximated by the hydrogenic levels with $Z = 1$.

5. The ground state of neutral beryllium ($Z = 4$) is $1s^2 2s^2$. Use the electron screening model to predict the energies of the following excited states:
   a) $1s^2 2s^1 3p^1$ (measured −2.02 eV)
   b) $1s^2 2s^1 4d^1$ (measured −0.90 eV)

The outer electron is screened by the three inner electrons, so $Z_{\text{eff}} = 4 - 3 = 1$. The expected energy of the $3p$ excitation is

$$E_n = (-13.6\text{ eV})\frac{Z_{\text{eff}}^2}{n^2} = (-13.6\text{ eV})\frac{1^2}{3^2} = -1.51\text{ eV}$$

and for $4d$

$$E_n = (-13.6\text{ eV})\frac{Z_{\text{eff}}^2}{n^2} = (-13.6\text{ eV})\frac{1^2}{4^2} = -0.85\text{ eV}$$

X-Rays

6. The minimum wavelength of the continuous X-ray spectra from an x-ray tube is $1.24 \cdot 10^{-11}\text{m}$. What is its working potential?

A photon with minimal wavelength occurs if the electron loses all its energy in one interaction with an atom (by Brehmsstrahlung). A photon with a wave length of $\lambda_{\min} = 1.24 \cdot 10^{-11}$ m $= 1.24 \cdot 10^{-2}$ nm has an energy

$$E = \frac{hc}{\lambda_{\min}} = \frac{1240\text{ eV nm}}{1.24 \cdot 10^{-2}\text{ nm}} = 10^5\text{ eV}.$$

The voltage needed to accelerate an electron to an energy of 10^5 eV is (by definition) $V_{AC} = 10^5$ V.

7. The K-absorption edge is the frequency at which, by the photoelectric effect, a photon has enough energy to release an electron from the K-shell (i.e., n = 1) of an atom. That means that the absorption of electromagnetic radiation (X-rays) increases when the frequency exceeds the frequency of the K-absorption edge. Similarly, the L-absorption edge is the frequency at which a photon can release an electron from the L-shell (n = 2). (The two X-ray contrast media iodine and barium have ideal K-shell binding energies for absorption of X-rays, 33.2 keV and 37.4 keV, respectively, which is close to the mean energy of most diagnostic X-ray beams.)

   a) What is the relation between the frequency of the K-absorption edge and the binding energy of an electron in the K-shell of the atom?
   b) The K-absorption edge of lead is at 0.14 Å. The wavelengths of the spectral lines of its K-series are 0.167 Å(K_α), 0.146 Å(K_β), and 0.142 Å(K_γ), respectively. Draw the X-ray energy level diagram for lead according to these values.
   c) Calculate the minimum energy required to activate the L series and the wavelength of the L line of lead.

a) The K absorption edge is the frequency (or wavelength) at which, by the photoelectric effect, a photon has enough energy to release an electron from the K-shell (i.e., $n = 1$). Hence the energy of a photon with the frequency of the K absorption edge equals the binding energy of an electron in the K-shell (i.e., with the lowest possible energy).

b) The K-absorption edge of lead at 0.14 Å=0.014 nm corresponds to an energy

$$E = \frac{hc}{\lambda_{\mathrm{K_{abs}}}} = \frac{1240 \text{ eV nm}}{0.014 \text{ nm}} = 88.6 \text{ keV}.$$

Hence the lowest energy level is $E_1 = -88.6$ keV.

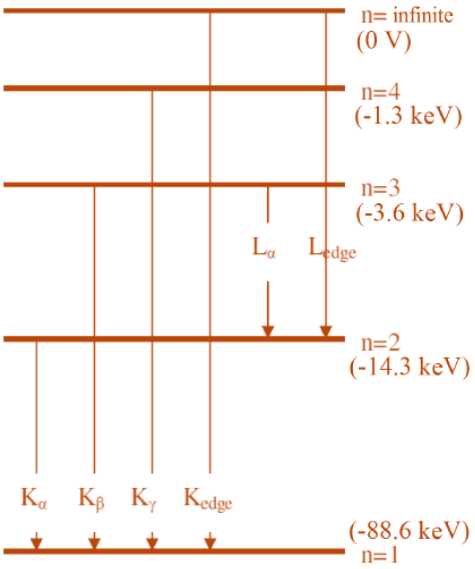


Figure 1: The X-ray energy diagram of exercise 6

The K_α series has a wavelength of 0.167 Å which corresponds to a photon energy of 74.3 keV. Hence $E_2 = E_1 + 74.3 = -14.3$ keV. For the K_β series (0.146 Å) the photon energy is 84.9 keV, which yields $E_3 = E_1 + 84.9 = -3.6$ keV. For the K_γ series (0.142 Å) the photon energy is 87.3 keV, which yields $E_4 = E_1 + 87.3 = -1.3$ keV. See Figure 1 for the corresponding X-ray energy diagram.

c) To release an electron from the L-shell an energy $E_{\mathrm{Ledge}} = -E_2 = 14.3$ keV is needed. The L_α X-ray has an energy $E_3 - E_2 = 10.7$ keV, which corresponds to a wavelength 0.116 nm.

8. An X-ray tube with aluminum as its target can produce a continuous X-ray spectrum with a short-wave limit of 5.00 Å. Is it possible to see the aluminum characteristic K_α X-ray in its spectrum? What is the case if its short wave limit is 10 Å? The K-absorption edge of aluminum is 1.559 keV.

To see the K_α line first an electron must be removed from the lowest (K) energy level. That requires electrons with an energy equal to the K absorption edge, which is for aluminium at 1.559 keV. If the short wave limit of the tube is 5 Å, which corresponds to a photon energy of 2480 eV=2.48 keV, there must be electrons (to produce the photons by Brehmsstrahlung) also with an energy of 2.48 keV. Hence in that case the K_α line can be seen. However, if the short wave limit of the X-ray tube is 10 Å, which corresponds to 1.24 keV, the K_α line cannot be seen.

9. An atom in its ground level absorbs a photon with energy equal to the K-absorption edge. Does absorbing this photon ionize this atom? Explain.

Yes, the energy of K absorption edge is equal to the energy to release an electron from the K-shell, which is the ionization energy of the atom.